#9 · 4.0% of marks

Ch 2 — Risk Management

Short chapter — learn every list. SM TYK theory Qs: **significance of VaR** (3.2) · **financial risk from different perspectives** (2). Numerical: VaR for a single share / two-share portfolio.

1 Types of risk faced by an organisation (SM 1)

Risk	SM meaning & example
Strategic	*Company's strategy becomes less effective and it struggles to reach its goal — technology change, new competitor, shift in demand, input-cost rise.* ***Kodak*** *(built a digital camera in 1975, saw it as a threat, left behind) ·* ***Nokia*** *(late to touch screens → Samsung led) ·* ***Xerox*** *(positive — moved into laser printing)*
Compliance	*Complying with Companies Act 2013, SEBI etc.; non-compliance → fine / imprisonment. Real problem when entering a* ***new business line or geography*** *(cement company entering sugar in another state)*
Operational	***Internal****; failure to cope with day-to-day problems; relates to* ***people and process*** *(paying ₹1,00,000 instead of ₹10,000 → second checker or system flag)*
Financial	*Unexpected changes in prices, exchange rates, credit rating, interest rates; political risk included as it leads to country risk*

2 Categories of financial risk (SM 1.4)

- → ***Counter-party****: non-honouring of obligations (goods not delivered after payment, loan / interest not repaid) — includes* ***credit risk***
- → ***Political****: faced by overseas investors — (i) confiscation / destruction of property (ii) rationing of remittance (iii) restriction on currency conversion (iv) restriction on borrowings (v) invalidation of patents (vi) price control*
- → ***Interest rate****: rate changes alter assets & liabilities; most important for* ***banks*** *(earn the spread). Inherent in both fixed and floating rates — floating borrower suffers when rates rise; fixed borrower pays comparatively more when floating rates fall*
- → ***Currency****: cash flows change with exchange rates, adversely or favourably — ₹ depreciation helps an exporter (****Infosys****), hurts an importer (****IOC****)*
- → ***Liquidity****: inability to meet liabilities when due — inadequate cash or mismatch in timing; common in banks (maturity mismatch)*

3 Evaluation of financial risk — 3 views (SM 2) — TYK Q2

View	SM
Stakeholders	*Equity holders see* ***financial gearing*** *as risk (last in winding up); lenders see high gearing as risk of default on interest & principal*
Company	*Excessive borrowing, or lending to someone who defaults, can force* ***liquidation***
Government	*Failure of a bank (****Lehman Brothers****) or downgrade of a financial institution spreads distrust; includes* ***wilful defaulters****; extends to* ***sovereign debt crisis***

TYK THEORY Q1 + NUMERICAL

④ Value at Risk (SM 3)

*Measure of investment risk: under normal market conditions over a set period (say one day), **how much an investment might lose** — portfolio, capital investment or forex. Answers: (i) what is the **worst case**? (ii) what will be the **loss**? First applied **1922 at NYSE**; entered finance in the **1990s**; now the world's most widely used measure of financial risk.*

Features (SM 3.1)

*(i) **Components**: time period · confidence level (generally **95% / 99%**) · loss in % or amount (ii) **statistical** tool based on standard deviation (iii) any **time horizon** — day, week, month (iv) **probability** of maximum loss, assuming normal distribution (v) **risk control** by setting maximum-loss limits (vi) **Z score** = number of σ from the mean; Z × σ = VaR*

Application / significance (SM 3.2) — TYK Q1

*(a) measure **maximum possible loss** on a portfolio or trading position (b) **benchmark** for performance measurement of operations / trading (c) fix **limits for front-office** dealers in treasury (d) help management decide **trading strategies** (e) tool for **ALM**, especially in banks*

VaR = Z × σ (in ₹) × √t · Z = **2.33** at 99%, **1.65** at 95% (one-tailed) · $\sigma_{t\ days} = \sigma_{1\ day} \times \sqrt{t}$

SM working	Answer
SM 3.3: ₹2 cr in X Ltd, daily σ 2%, 252 trading days, 99%	σ = ₹4 lakh → 1-day VaR = 4 × 2.33 = **₹9.32 lakh**; 10-day = √10 × 9.32 = **₹29.47 lakh**
TYK 1: ₹200 lakh each in XYZ & ABC, daily σ 1% each, r = 0.3, 10-day 99%	Daily σ each ₹2 lakh; variance = $2^2 + 2^2 + 2 \times 0.3 \times 2 \times 2 = 10.4$ → σ = **₹3.22 lakh** (or 0.80623% of ₹400 lakh); 10-day σ = 3.22 × √10 = **₹10.18 lakh**; VaR = 2.33 × 10.18 = **₹23.72 lakh**

⚠ TRAP

*Scale σ by **√t**, never by t. Use portfolio σ with the correlation term — two 1% shares with r = 0.3 give 0.806%, not 1%. 252 days is only needed to annualise.*

⑤ Identifying & managing financial risk (SM 4)

Counter-party risk (4.1)

***Hints (5)**: (a) failure to obtain resources to complete the project (b) regulatory restrictions (c) hostile action of a foreign government (d) let down by a third party (e) insolvency. **Management (6)**: (1) **due diligence** before dealing (2) don't over-commit to one entity / group (3) know exposure limits (4) review credit limits & approval procedure regularly (5) rapid action on likely default (6) performance guarantees, insurance, other instruments.*

Political risk (4.2)

Identify (4): insistence on resident investors / labour · restriction on currency conversion · **expropriation** of foreign assets · price fixation

Assess country risk (4): political rankings in business magazines · macro-economic conditions · popularity & stability of government · advice from home-country **embassies**

Mitigate (4): **local sourcing** of materials & labour · **joint ventures** · **local financing** · **prior negotiations**

Interest rate risk (4.3) — identified from (6)

Monetary policy · government action (eg **demonetisation**) · economic growth · release of industrial data · foreign investment · stock-market changes. Management → Ch 12.

Currency risk (4.4) — parameters (6)

(1) **Government action** (Brexit → £ lowest since the 1980s) (2) **nominal interest rate** (IRP) (3) **inflation** (PPP) (4) natural calamities (5) war, coup, rebellion (6) change of government and its attitude to foreign investment. Management → Ch 10.

Mnemonic for currency parameters: **G-I-I-N-W-C** — Government, Interest, Inflation, Natural calamity, War, Change of government.

⑥ Correlation — how Ch 2 actually connects

Ch 2 idea	The actual link (app-written cross-reference; the Ch 2 side is SM)
VaR of a 2-share portfolio	**Ch 6**: $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2r\sigma_1\sigma_2$ — the same formula; lower r → lower σ_p → lower VaR. That is diversification expressed in rupees.
Interest rate risk	**Ch 12**: identified here, managed there (gap, FRA, caps, swaps); **Ch 5**: price vs reinvestment risk, duration.
Currency risk — IRP, PPP	**Ch 10**: the parity conditions forecast the rate; the hedges (forward, MMH, options) manage it; **Ch 11**: political-risk list = complexity list in international capital budgeting.
Political risk mitigation	**Ch 11**: local financing and JVs reduce blocked-fund and expropriation risk; certainty-equivalent cash flows rather than a higher discount rate.
Operational risk, limits	**Ch 9**: Barings (back office run by the trader), Orange County — lessons = separate front / middle / back office, risk limits, VaR + stress tests.
Financial gearing as risk	**FM / Ch 14**: high gearing raises K_e; LBO targets carry the most gearing risk; **AAA**: SA 570 going-concern indicators (defaults, negative net worth).